

Weather-Driven Electricity Demand Prediction

CPSC 482 Final Project

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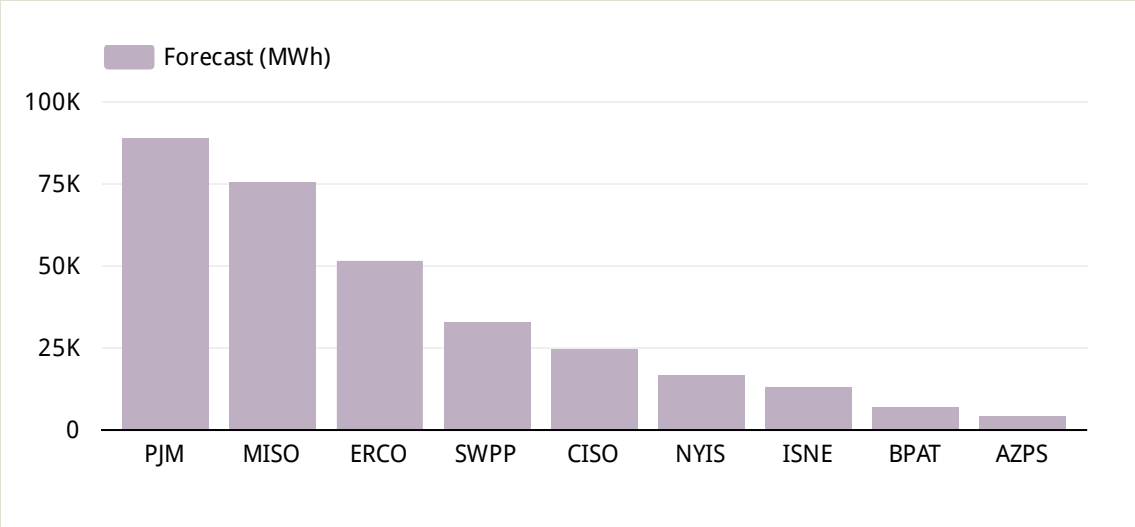
Filter by Grid Region

Model Performance Comparison

	Model	Mean Average Error ▾	Mean Squared Error	R2 Score
1.	Linear Regression	4,557	49,755,419.45	0.95
2.	Boosted Tree	1,919.34	9,296,965.2	0.99
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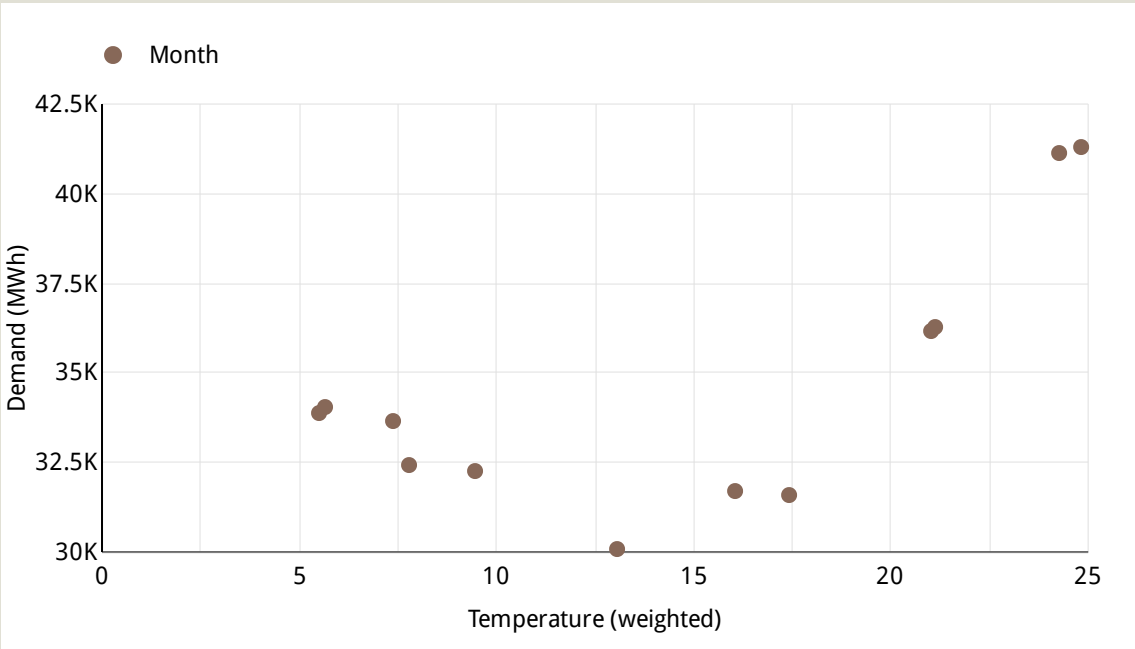
Evaluated on 2022 data. Lower MAE/MSE and higher R² indicate better performance

Average Electricity Demand by Grid Region (MWh)



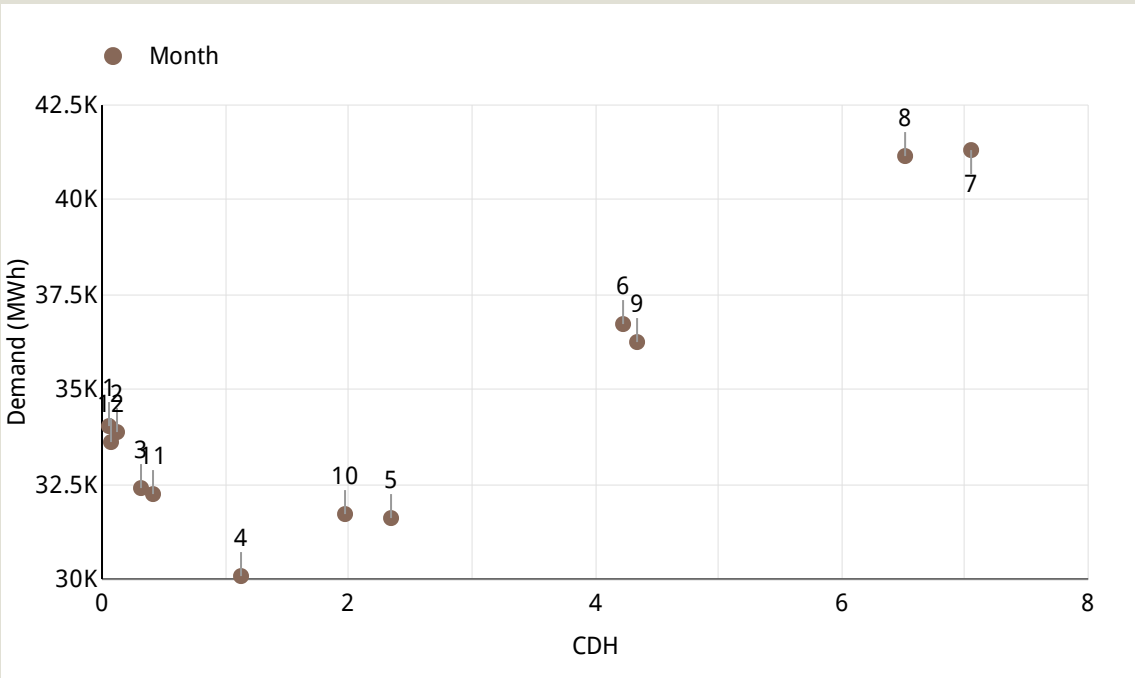
PJM and MISO are the largest grid regions by demand volume, showing how they are more dense population centers.

Temperature vs. Electricity Demand



Population weighted regional temperature plotted against hourly demand. Each point represents a balancing authority

Cooling Degree Hours vs. Electricity Demand



CDH shows temperatures that are above threshold, influencing air conditioning load. Higher CDH correlates with higher summer demand.

Solar Energy vs. Electricity Demand

